

Irradiation effects of a 10 MeV neutron beam on a Nd–Fe–B permanent magnet

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ABSTRACT

The effects of irradiation of a Nd–Fe–B permanent magnet by fast neutrons was investigated. The decrease in measured magnetic flux density at the center of the magnets were 0.6%, 6.9%, 25.2% and 47.3% after continuous irradiation of 1.1 kGy, 3.7 kGy, 5.6 kGy and 7.4 kGy, respectively. On the other hand, the decrease due to non-continuous irradiation, in which the magnet was first irradiated at 3.7 kGy, then irradiated again at 3.7 kGy nine months later, was 14% smaller than that of continuous irradiation, even for the same total dose. The temperature coefficient of the magnetization did not change with irradiation. Some radioactive materials, such as ^{147}Nd , ^{151}Pm , and ^{54}Mn , were detected in the magnet after irradiation.

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1. Introduction

Nd–Fe–B permanent magnets have special characteristics concerning compactness, such that they require no power supply and no cooling system. These magnet materials are characterized by a high remanent flux density of 1.2–1.5 T, and an energy product of up to 400 kJ/m³. Therefore, many kinds of applications in the accelerator field such as magnetic focusing equipment [1], ECR ion sources [2], insertion devices for synchrotron radiation [3] and cyclotrons for AMS [4] have been reported.

At NIRS (National Institute of Radiological Sciences), a compact cyclotron has been studied for use in the production of positron emitter radioisotopes at low operational cost. Here, Nd–Fe–B permanent magnets will be employed as the main-magnets of the cyclotron. In this case, a quantity of high-energy neutrons are supposed to come from internal RI-production targets, extraction devices, beam shutters and so on. Thus, demagnetization of the permanent magnets due to high-energy beam irradiation must be studied.

Several studies concerning the demagnetization of Nd–Fe–B permanent magnets with irradiation have been reported, as listed in Table 1 [5–9]. It seems that demagnetization strongly depends

on the kind of radiation. Alderman et al. carried out X ray, ^{60}Co gamma ray, ^{252}Cf thermal and fast neutron irradiation experiments, and reported that only fast neutron irradiation damaged the magnetization [10]. In the present work, we focused on demagnetization with fast neutron irradiation with a central energy of 10 MeV. The temperature dependences of the magnetization and the residual activities in the magnets were also investigated to study the applicability of Nd–Fe–B permanent magnets for a cyclotron.

A Nd–Fe–B permanent magnet, NEOMAX-39SH, was used in the experiment. The main specifications of the magnet are listed in Table 2. A sample from the permanent magnet was formed into a rectangular shape of 25 mm × 25 mm and 5 mm thickness, and the direction of magnetization was parallel to the short axis.

2. Experimental setup

2.1. Irradiation system

A fast neutron flux for irradiating the magnet was produced by a method involving a deuteron–Be reaction system, which was equipped in one of the vertical beam lines at the NIRS-cyclotron facility [11]. During the reaction, the measured gamma ray dose was less than 5% of the total dose [12]. Therefore, the gamma ray dose was too small to induce any demagnetization in the sample

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Table 1

Typical references of demagnetization of Nd-Fe-B permanent magnets with irradiation.

Kind of radiation	Energy or source	Demagnetization (%)	Dose (kGy)	Ref.
Gamma-ray	^{60}Co	0.5	2800	[6]
Electron	20 MeV	9	2600	[6]
Electron	85 MeV	14	4500	[9]
Proton	500 MeV	50	40	[8]
Proton	200 MeV	20	2000	[7]
Neutron	Reactor	50	$3.2 \times 10^{16} \text{ n/cm}^2$	[5]

Table 2

Main specifications of the Nd-Fe-B permanent magnet.

Material code	NEOMAX-39SH
Max BH products (kJ m^{-3})	319
Remanence B_r (T)	1.29
Coercivity H_{CB} (kA m^{-1})	1003
Coercivity H_{CJ} (kA m^{-1})	1671
Temp. coeff. of B_r (%)	-0.11

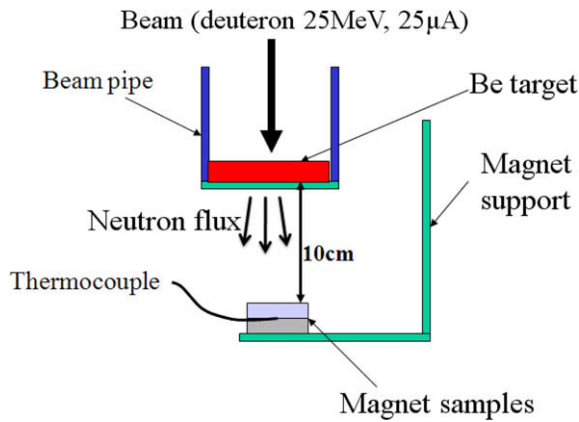


Fig. 1. Schematic view of the sample magnets and irradiation system. Two samples were set in a pile. The magnetization direction was parallel to the beam direction.

magnet. A neutron flux, having a central energy of about 10 MeV was produced by a 25 MeV deuteron beam on a water cooled Be target of 3 mm thickness. Usually, the neutron flux is used for biological experiments, and a sample is irradiated at 2 m downstream from a Be target. At that position, a neutron dose rate of 0.154 Gy/min was measured using an ion chamber when the incident deuteron beam current was 25 μA .

Takada et al. measured the neutron energy spectrum at the same beam line and position, the results are shown in Fig. 2 in Ref. [12]. The total neutron flux of $1.16 \times 10^{11} \text{ n}/(\mu\text{Cs})$ was evaluated by integrating the numerical data of the neutron energy up to 29 MeV.

In this experiment, the magnet was set 10 cm downstream from the Be target, as shown in Fig. 1. Therefore, the neutron flux rate was estimated to be $2.90 \times 10^{10} \text{ n}/(\text{cm}^2\text{s})$ and a dose rate of 61.6 Gy/min was estimated at the same beam current.

Since the distance between the Be target and the magnet was very close, the spatial distribution of the neutrons at the irradiation position was measured using an aluminum-activation method [13]. In this method, radioactive isotopes of ^{27}Mg were produced from the $^{27}\text{Al}(n,p)$ reaction by neutron irradiation on an aluminum plate. After irradiation, the aluminum plate was attached to an imaging plate to be exposed with 0.844 and 1.014 MeV photons from ^{27}Mg . This technique had an advantage that fast neutrons at more than 2.5 MeV were measured.

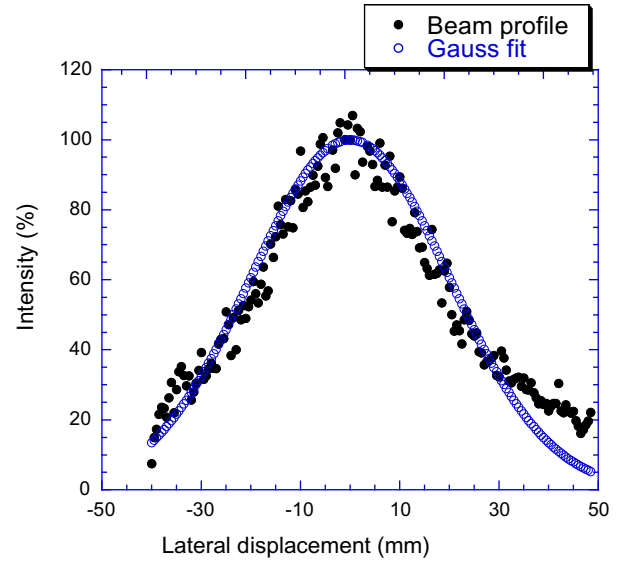


Fig. 2. Measured density distribution of the neutron flux on the irradiation position together with a Gaussian fit.

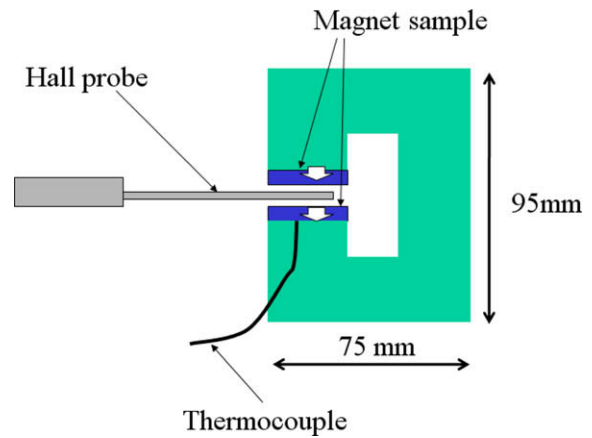


Fig. 3. Magnetic field measurement system consisting of a C-type yoke having a thickness of 25 mm. The size of the sample magnet was 25 mm $\times$ 25 mm square and 5 mm thick. Arrows indicate the direction of magnetization.

Fig. 2 shows the measured profile of the neutron flux distribution at the position of the irradiation together with a Gaussian fitted curve. The neutron flux at the magnet edge is 80% of that at the center.

2.2. Magnetic field measurement system

A simple test stand consisting of a C-type iron yoke was prepared to measure the magnetic field distribution of the magnets, as shown in Fig. 3. A pair of magnets can be easily mounted on, and then taken off from the yoke in order to measure the magnetic field strength before and after neutron irradiation.

In this layout, the magnetic flux density (B_g) in the magnet gap is given by the remanent flux density (B_r) of the permanent magnet, as follows:

$$B_g = k \cdot B_r \cdot L / (L + d) \quad (1)$$

where k is a constant determined by the structures of yoke and permanent magnets, L is the thickness of the permanent magnets ($L = 5 \text{ mm} \times 2$), and d is the gap distance ($d = 5 \text{ mm}$). Eq. (1) is valid if the relative magnetic flux density distribution at the magnet gap

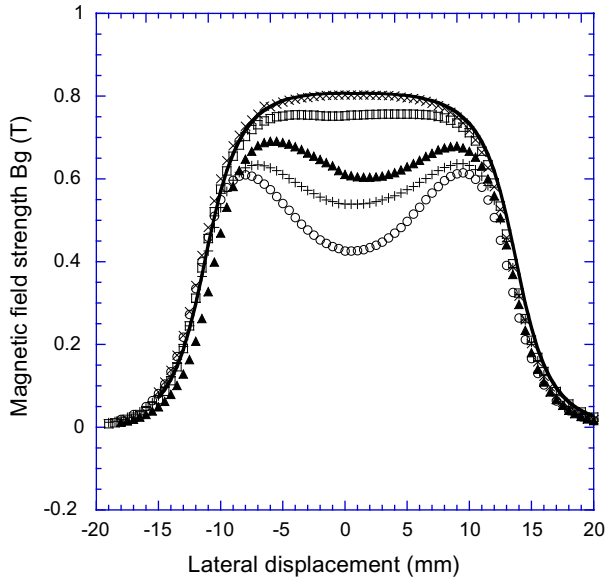


Fig. 4. Comparison of the magnetic field distributions before and after neutron irradiation under different irradiation cases: line, before irradiation; $\times$, 0.3 h; $\square$, 1 h; $\blacktriangle$, 1.5 h; $+$, 1 h + 1 h and $\circ$, 2 h.

does not depend on the magnitude of B_r . Thus, the change in B_r (δB_r) is directly connected to the change in B_g (δB_g) as

$$(\delta B_r / B_r) = (\delta B_g / B_g) \quad (2)$$

In the present study, the magnitude of $(\delta B_r / B_r)$ is called the demagnetization. The magnetic field on the median plane of the gap was measured using a Hall probe, where the temperature coefficient of the Hall probe was $-0.075\%/K$ at around 298 K. The size of the Hall unit was $0.07 \text{ mm} \times 0.07 \text{ mm}$, which is small enough to measure the magnetic field with the required spatial resolution.

A field map on the median plane was measured by using an X–Y table having a setting resolution of better than $1 \mu\text{m}$. A magnetic field measurement was performed at $295 \text{ K} \pm 1 \text{ K}$, that is, at room temperature.

The permanent magnet was set in the yoke before irradiation, and the magnetic field distribution of the median plane of the gap was measured. The result is shown later in Fig. 8 together with numerical calculations using the 3-D field simulation code, TOSCA [14], for the same layout of the measurement system. The value of B_r used for the calculation was 1.29 T, as shown on Table 2. The measurements were well reproduced by calculations within errors of 1%. Thus, the constant k was calculated to be 0.94, and the remanent flux density before irradiation is compatible with the magnet's specifications.

3. Experiments

3.1. Demagnetization

A pair of sample magnets were united in a pile, and set at the irradiation position, as shown in Fig. 1. The magnetization direction was parallel to the axis of the beam line.

In this experiment, the neutron irradiation dose was expressed as a function of the irradiation time, because the dose could not be measured directly in the flux range of interest. Continuous irradiations were performed on each pair of samples with different irradiation times of 0.3, 1.0, 1.5 and 2 h continuously, which are equivalent to total doses of 1.1×10^3 , 3.7×10^3 , 5.6×10^3 , and $7.4 \times 10^3 \text{ Gy}$, respectively. Furthermore, non-continuous irradiation was carried out in which the magnet was irradiated for 1 h

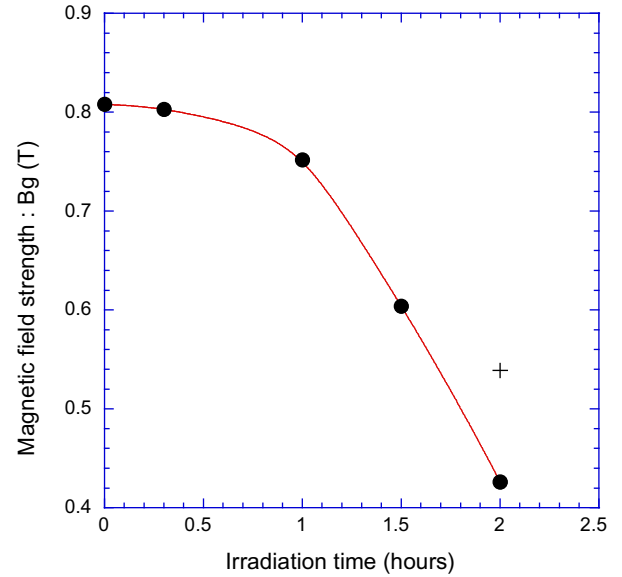


Fig. 5. Dose dependence of the magnetic field strength at the magnet gap center in the measurement system. Round symbols are the case of continuous irradiation, and the + symbol represents non-continuous irradiation.

Table 3

Induced RI and activities.

RI	HL (min)	Energy (keV)	B_q (g)
^{147}Nd	11.98 d	91.2	1705
^{139}Ce	133.7 d	165	145
^{151}Pm	28 h	340	5610
^{54}Mn	312.5d	835	1390
^{48}Sc	43.7 h	1037	9.5
^{59}Fe	44.6 d	1292	189
^{60}Co	5.271 y	1333	43.4
^{52}Mn	5.59 d	1434	37.5
Total B_q (g)			9130

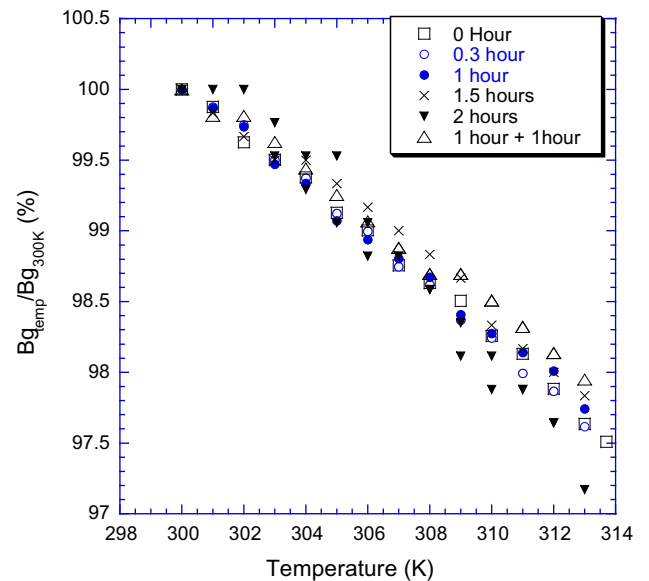


Fig. 6. Temperature dependence of the magnetic flux density. Each measured data was normalized by B_g at 300 K.

initially, and then another 1 h nine months later. Due to radioactivity of the samples, a cooling time of two weeks was allowed after

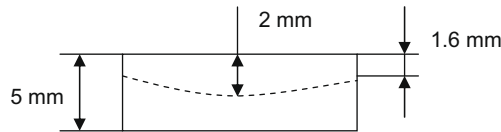


Fig. 7. Schematic illustration of scraped off the pole face of a pair of sample magnets for a TOSCA simulation.

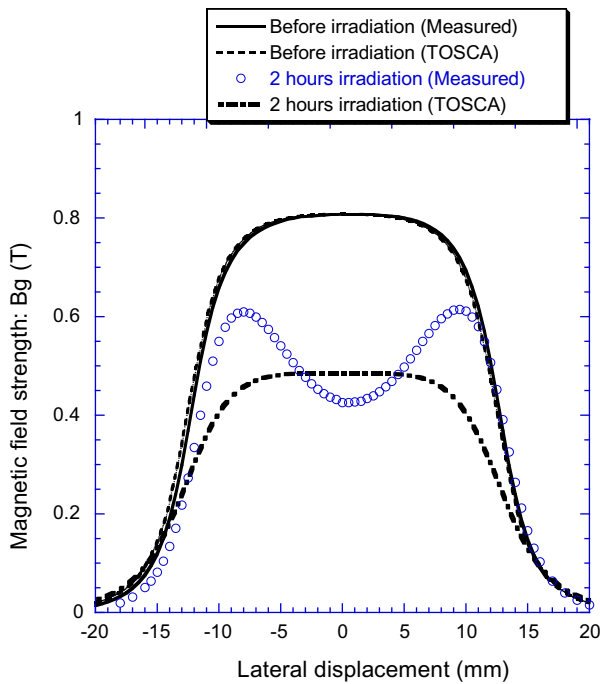


Fig. 8. Measured and calculated magnetic flux density distributions after 2 h of irradiation and before irradiation.

each irradiation. The pair of sample magnets were set in a test stand, and the magnetic field distribution was measured on the median plane. The results are shown in Figs. 4 and 5, where a non-continuous irradiation is indicated by the symbol (+).

The spatial magnetic field distributions were changed significantly, depending on the dose. Thus, Eq. (2) was not applied to determine the demagnetization from the present experiment. Therefore, we call the ratio $\delta B_g/B_g$ at the center of the median plane the central field decrement. It seems from the differences among the field distributions that the demagnetizations are not proportional to the neutron dose. On the other hand, the central field decrement of non-continuous irradiation was 14% smaller than that of the continuous irradiation for the same total dose.

During irradiation, the temperature of the sample magnets was also remotely measured by a thermocouple. The maximum temperature rise of 12 °C was measured during 2 h of irradiation at an environmental temperature of 24 °C, which is much lower than the Curie point of 585 K.

The spatial magnetic field distributions of the irradiated sample magnets were measured again after nine months and two years. The results showed that all of the magnets stayed as they were.

3.2. Residual activities and temperature dependence

Long-lived residual activity produced by neutron irradiation was measured two weeks after irradiation with a Germanium detector (EG & G GMX-30200). The detector had a window whose diameter was 50 mm, and was covered with a 3 mm PMMA sheet. Long-lived isotopes of ^{147}Nd , ^{59}Fe , ^{139}Ce , and ^{54}Mn were detected

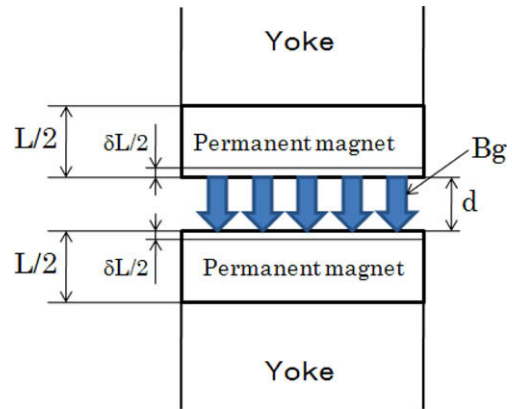


Fig. A1. Schematic illustration of a magnet gap and permanent magnets.

in a counting time of 10^5 s. The main nuclides are listed in Table 3, together with their half-lives. The doses of these activities, however, were reduced to 0.3 $\mu\text{Sv/h}$ two weeks after irradiation on the magnet sample which had undergone 2 h irradiation.

The temperature dependence of the magnetic field was measured both before and after irradiation between temperatures of 298 and 314 K with a step of 1 K. The results are shown in Fig. 6. The temperature coefficient of the magnetization is compatible with the magnet's specifications, and did not change with irradiation.

4. Summary and discussions

The central field decrements with continuous irradiations were 0.6%, 6.9%, 15.2% and 47.3% for doses of 1.1 kGy, 3.7 kGy, 5.6 kGy and 7.4 kGy, respectively. The central field decrement for non-continuous irradiation was 14% smaller than that for continuous irradiation, even for a same total dose of 7.4 kGy.

In order to understand the demagnetization phenomenon, numerical calculations of the magnetic field were carried out using the TOSCA code for 2 h of irradiation. In this simulation, it was assumed that the demagnetization is proportional to the density distribution of the neutron flux (see Fig. 2) and is equivalent to the lack of a pair of a magnet pole face, as shown in Fig. 7 which is proportional to the density distribution of the neutron flux (see Appendix 1). The results of the calculations were quite different from the measurements, as shown in Fig. 8. Thus, the demagnetization was not proportional to the neutron dose distribution, but strongly depended on the dose rate.

In order to determine whether the Nd–Fe–B permanent magnet can be applied to the main-magnet of a compact cyclotron, a neutron environment was measured at the NIRS-cyclotron facility using CR-39 with a polyethylene proton radiator. The measured flux rate was 777.5 $\text{n}/(\text{cm}^2 \mu\text{A min})$ when the cyclotron accelerated a proton beam of 18 MeV. Assuming that the compact cyclotron operates with 100 μA , 24 h a day, the neutron flux around the cyclotron was calculated to be $1.12 \times 10^8 \text{ n/cm}^2$. On the other hand, in the present experiments, the central field decrement was 0.6% for a neutron flux of $3.13 \times 10^{13} \text{ n/cm}^2$ (0.3 h of irradiation). Therefore, the demagnetization of the Nd–Fe–B magnet to be expected from a 24 h a day operation is below $2.15 \times 10^{-6}\%$.

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Appendix 1. Relation between a demagnetization and thicknesses of permanent magnets

Fig. A1 shows the details of the permanent magnets and the magnet gap that was used for field measurements (see Fig. 3). In the figure, $L/2$ is the initial thickness of the permanent magnet and $\delta L/2$ means the part scraped away from the initial magnet. The initial magnetic field in the gap is given by Eq. (1) as

$$B_g = k \cdot B_r \cdot L / (L + d) \quad (1)$$

We now consider the field strength when the thickness of the permanent magnet becomes $(L - \delta L)/2$, as shown in Fig. A1. We can then see from the figure that the magnet gap becomes $d + \delta L$. If the condition of $\delta L \ll L$, d is valid, and thus the change of k is small, the new magnetic field B'_g is given by the following equation:

$$B'_g = k \cdot B_r (L - \delta L) / (L - \delta L) + (d + \delta L) = k \cdot B_r (L - \delta L) / (L + d) \quad (A1)$$

Where Eq. (A1) is valid if the relative magnetic flux density distribution at the magnet gap does not depend on the thickness of the permanent magnet.

Thus, the field decrement δB_{gl} due to the lack of a permanent magnet is given by

$$\delta B_{gl} = B_g - B'_g = k \cdot B_r \cdot \delta L / (L + d)$$

Therefore, the relative field decrement, $\delta B_{gl}/B_g$ is given by

$$\delta B_{gl}/B_g = \delta L/L \quad (A2)$$

When the value of $\delta B_{gl}/B_g$ is equal to $\delta B_g/B_g$ in Eq. (2), we obtain

$$\delta B_r/B_r = \delta L/L \quad (A3)$$

Namely, the field change due to a demagnetization is equivalent to the change caused by a decrement in the thickness of the permanent magnet. The above statement, however, includes several assumptions. Thus, Eq. (A3) is approximately valid.

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